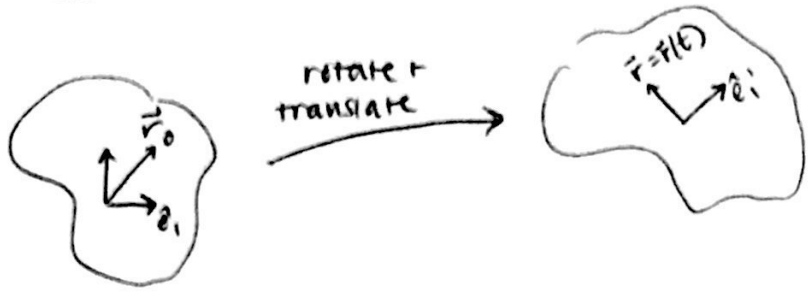


Rotations:

Recall



$$\vec{r} = \underline{R} \cdot \vec{r}_0$$

$$[\vec{r}]_{\beta} = [\underline{R}]_{\beta} \cdot [\vec{r}_0]_{\gamma}$$

Fact: $[\vec{r}]_{\beta} = [\vec{r}_0]_{\gamma}$

2 interpretations of $\vec{r} = \underline{R} \vec{r}_0 \rightarrow$ $\begin{cases} r_0 = \text{components of } \vec{r} \text{ in fixed frame before rotation} \\ \vec{r}_0 = \text{components of } \vec{r} \text{ in } \beta \text{ now} \end{cases}$

Compound Rotations

Rotate then rotate

rotate $\underline{R}_1 \vec{r}_0$ then rotate $\underline{R}_2 \underline{R}_1 \vec{r}_0 \neq \underline{R}_1 \underline{R}_2 \vec{r}_0$

Rotations do not commute

Enter Angles:

$$R_{ij} = 9 \# S$$

columns are unit vectors w/ magnitude 1 $\Rightarrow$ 3 constraints

columns are mutually orthogonal $\Rightarrow$ 3 constraints

3 independent numbers

or

Axis angle - $4 \# S$ $|\hat{n}| = 1$ 1 constraint $\Rightarrow$ 3 independent $\# S$

Ideal: $\begin{bmatrix} n_1 \theta \\ n_2 \theta \\ n_3 \theta \end{bmatrix}$ — from axis angle

Idea 2: Euler Angle

Axis angle

, try to rotate

$$\text{ex) } \underline{R}_{\text{about z axis}} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

24 choices for Euler Angle

angle of rotation about 3 successive axis

$\underline{R}_a, \underline{R}_b, \underline{R}_c$ using old axes

$$\underline{R}_c \underline{R}_b \underline{R}_a$$

opposite

or

$$\underline{R}_a, \underline{R}_b, \underline{R}_c$$

new axis

$$\underline{R}_a \underline{R}_b \underline{R}_c$$

$$\left[\begin{array}{c} \text{A then B then C} \\ \uparrow \quad \quad \uparrow \\ \text{original} \quad \text{original} \\ \hline = \text{C then new B then new A} \end{array} \right]$$

Small Angles

Axis $\hat{n}$ & small angle $\Delta\theta$

$$\Delta\theta \ll 1$$

$$\sin\Delta\theta \sim \Delta\theta$$

$$\cos\Delta\theta \sim 1$$

$$\underline{R} = \underset{0}{\cancel{\hat{n}\hat{n}}} + \cos\theta \left(\underset{1}{\underline{1}} - \underset{0}{\cancel{\hat{n}\hat{n}}} \right) + \cancel{\hat{n}}(\hat{n})\sin\theta$$

$$\underline{R}(\hat{n}, \Delta\theta \ll 1) \approx \underline{1} + \Delta\theta \hat{n}$$

$$\underline{R}(\hat{n}, \Delta\theta) \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \Delta\theta \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix}$$

$$\underline{\underline{R}} \cdot \underline{\underline{r}} = \underline{\underline{r}} + \Delta\theta \hat{n} \times \underline{\underline{r}}$$

$$\frac{d\underline{\underline{r}}}{dt} = \underbrace{\frac{\Delta\theta}{\Delta t}}_{\underline{\underline{\omega}}} \hat{n} \times \underline{\underline{r}}$$

$$\boxed{\frac{d\underline{\underline{r}}}{dt} = \underline{\underline{\omega}} \times \underline{\underline{r}}} = \underline{\underline{\omega}} \cdot \underline{\underline{r}} \rightarrow \mathcal{L}(\underline{\underline{\omega}})$$

$$\underline{\underline{r}} = \underline{\underline{R}} \cdot \underline{\underline{r}}_0$$

↑
constant

$$\rightarrow \frac{d\underline{\underline{r}}}{dt} = \underline{\underline{\dot{R}}} \cdot \underline{\underline{r}}_0$$

$$= \underline{\underline{\omega}} \cdot \underline{\underline{r}}$$

$$= \underline{\underline{\omega}} \cdot (\underline{\underline{R}} \cdot \underline{\underline{r}}_0)$$

$$\underline{\underline{\dot{R}}} = \underline{\underline{\omega}} \cdot \underline{\underline{R}}$$

$$\underline{\underline{\omega}} = \underline{\underline{\dot{R}}} \cdot \underline{\underline{R}}^T$$

$$\underline{\underline{\dot{r}}} = \underline{\underline{\dot{r}}}$$

$$\underline{\underline{\omega}} \cdot (\underline{\underline{R}} \cdot \underline{\underline{r}}_0) = \underline{\underline{\dot{R}}} \cdot \underline{\underline{r}}_0 \text{ for all } \underline{\underline{r}}_0$$

Look at dyads:

$$\underline{\underline{R}} = \hat{e}_i' \hat{e}_i$$

$$\underline{\underline{\dot{R}}} = \dot{\hat{e}}_i' \hat{e}_i$$

$$= (\underline{\underline{\omega}} \times \hat{e}_i') \hat{e}_i$$

$$= \underline{\underline{\omega}} \cdot \hat{e}_i' \hat{e}_i$$

$$\underline{\underline{\dot{R}}} = \underline{\underline{\omega}} \cdot \underline{\underline{R}}$$